| Centre Number | Examination Number |
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04300294

# EXAMINATIONS COUNCIL OF ZAMBIA

**Examination for General Certificate of Education Ordinary Level**

# Design and Technology 6045/1

## Paper 1 (Theory & Design)

**Tuesday** **1 AUGUST 2023**

**Additional Materials**
A3 Drawing Paper
Drawing instruments

**Time: 2 hours 30 minutes** **Marks: 110**

### Instructions to Candidates

1 Write the **centre number** and your **examination number** on **every page** of this question paper and on the **separate Answer Booklet** provided.

2 If working is needed for any question, it must be shown in the spaces provided in the question paper and Answer Booklet.

3 There are **two** sections in this paper, **A** and **B**.

**(i)** Answer **all** questions in Section A in the spaces provided in this question paper.

**(ii)** Section B has **three** parts **I**, **II** and **III**; Part **I** is compulsory.

**(iii)** Choose **one** question from Part **II** and write your answers in the separate Answer Booklet provided.

**(iv)** Choose **one** question from Part **III** and use the A3 drawing paper (Insert) provided.

**(v)** Enter the numbers of questions you have answered from Section B Parts **II** and **III** in the grid provided.

4 Where a freehand sketch is required, it must be drawn to half full size with good linework.

5 At the end of the examination, fasten the Answer Booklet and drawing paper used securely to the question paper.

6 Use your own discretion where measurements are not provided.

### Information for Candidates

1 The number of marks is given in brackets [ ] at the end of each question or part question.

2 Use of colour is allowed.

3 All dimensions are in millimetres.

4 Cell phones and calculators are **not allowed** in the examination room.

| *For Examiner's Use Only* | | | |
|---|---|---|---|
| | **Question** | **Mark Obtained** | **Examiner's Initials** |
| **Section A** | | | |
| **Section B** | | | |
| **Part I** | | | |
| **Part II** | | | |
| **Part III** | | | |
| **Total** | | | |

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| Centre Number | Examination Number |
|---|---|
| | |

**SECTION A: [30 MARKS]**

**Answer all the questions from this section in the spaces provided in this question paper.**
**If working is needed for any question, it must be shown in the space below that question.**

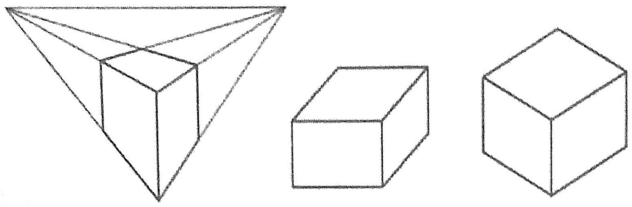
**1 (a)** State **three** safety precautions that should be taken when applying finishes.

..... [1]

..... [1]

..... [1]

**(b)** **Figure 1** shows various ways of representing a box in pictorial views.



**A** **B** **C**

**Figure 1**

**(i)** Identify each method of projection used in **Figure 1**.

**A** ..... [1]

**B** ..... [1]

**C** ..... [1]

**[Total: 6 marks]**

**2 (a)** Mention **two** factors that designers would consider when selecting materials for making an artefact.

..... [1]

..... [1]

**(b)** **Figure 2** shows a trophy awarded at the national JETS fair made from 1mm thick sheet of copper.




Material:Copper

**Figure 2**

**(i)** Name **one** suitable tool for cutting the 1mm thick copper material.

..... [1]

**(ii)** Suggest a suitable method of joining handles to the trophy

..... [1]

**(c)** **Figure 3** shows a picture frame made from softwood joined at 45°.

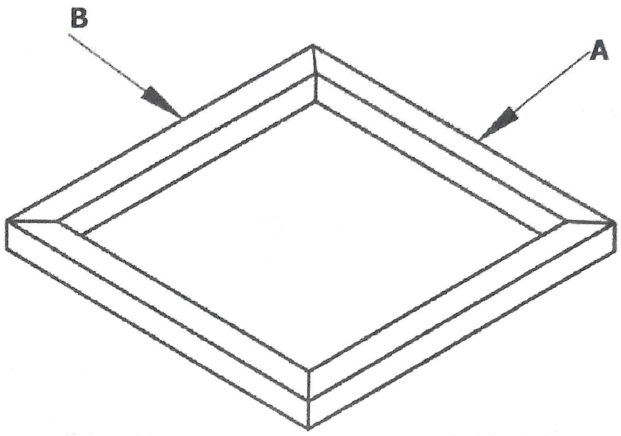



**Figure 3**

**(i)** Name an appropriate clamp that can be used when joining pieces **A** and **B**.

..... [1]

**(ii)** Suggest the type of glue you would use to join **A** and **B.**

..... [1]

**[Total: 6 marks]**

**3** **(a)** **Figure 4** shows components **A, B** and **C** in a simple electronic circuit.

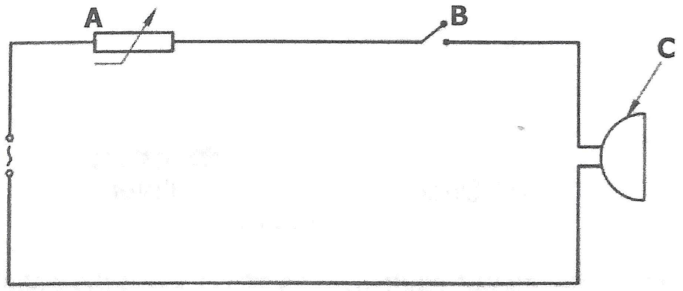



**Figure 4**

**(i)** Identify the component labelled **C.**

..... [1]

**(ii)** State the function of components **A** and **B.**

**A** ..... [1]

**B** ..... [1]

| Centre Number | Examination Number |
|---|---|
| | |

**(b)** **Figure 5** shows components used in generating electric power.

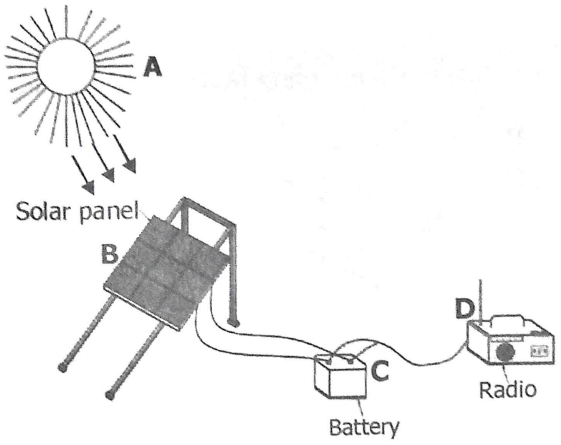



**Figure 5**

**(i)** State the energy changes taking place from

**A** to **B** ..... [1]

**B** to **C** ..... [1]

**(ii)** Suggest the advantage of using the method of generating power **B**.

..... [1]

**[Total: 6 marks]**

**4** **Figure 6** shows types of motions made by the wall clock and a piston.

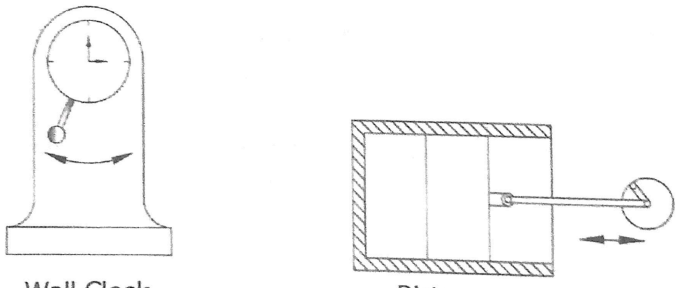



**Figure 6**

**(a)** Identify the motions made by the wall clock and the piston.

**(i)** Wall clock ..... [1]

**(ii)** Piston ..... [1]

**(b)** Mention the types of motions produced by the following devices:

**(i)** Fan ..... [1]

**(ii)** Tin snip ..... [1]

**(c)** Explain **two** characteristics of an entrepreneur.

**(i)** ..... [1]

**(ii)** ..... [1]

**[Total: 6 marks]**

**5** **(a)** **Table 1** shows drawings of hand tools. State the use of each tool.

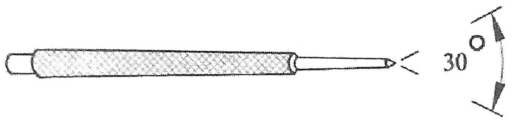
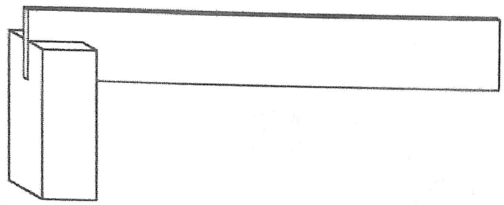
| Tool | Use |
| --- | --- |
|   | ..... ..... ..... [1] |
|  | ..... ..... ..... [1] |

**Table 1**

**(b)** The following items were made from metal, plastic and wood as the main materials. State the specific type of material from which each of the following items could be made:

**(i)** Knife

..... [1]

**(ii)** Shop sign

..... [1]

**(iii)** Chopping board

..... [1]

**(c)** **Figure 7** shows a fixing used in a workroom. When is the fixing in **Figure 7** used?

..... [1]

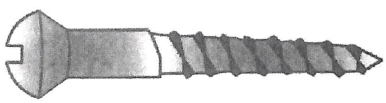


**Figure 7**

**[Total: 6 marks]**

| Centre Number | Examination Number |
|---|---|
| | |

**SECTION B: [80 MARKS]**

**There are three parts in this section; Part I, II and III. Part I is compulsory. Answer one question from Part II and one question from part III. Write your answers in the separate Answer Booklet provided.**

**Part I: Compulsory**

**6** Study **Figure 8** and answer the questions that follow.

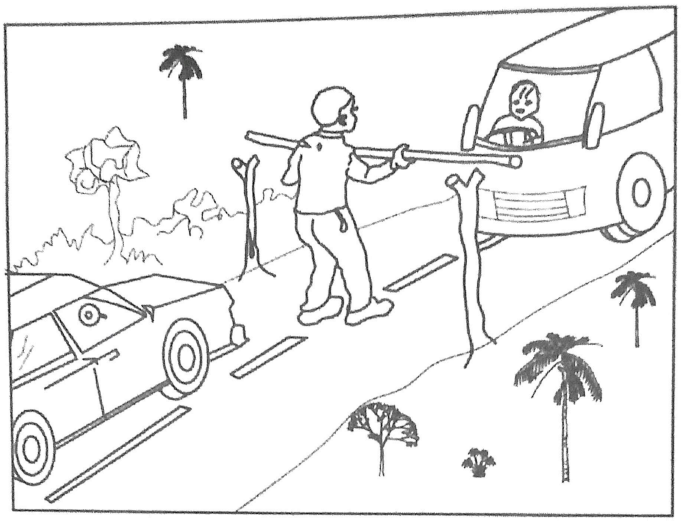


**Figure 8**

**(a)** **(i)** Write the **situation** obtaining in **Figure 8**. [2]

**(ii)** Identify the **problem** from the situation in **(a)(i)**. [2]

**(iii)** Write a **design brief** from the problem identified in **(a)(ii)**. [2]

**(b)** **(i)** Using annotations, sketch in pictorial view, one possible solution to the problem identified in the situation in **(a)(i)**. [10]

**(ii)** Mention any **two** methods of construction you would use in making the possible solution. [2]

**(iii)** In pictorial, **sketch** an exploded view of one method of construction in **(b)(ii)** [4]

**(c)** **(i)** Give **one** reason why exploded views are important in the design process. [2]

**(ii)** Suggest **one** safety consideration you would take during the realisation of the possible solution. [1]

**[Total: 25 marks]**

**Part II: Answer one question**

**7** **Figure 9** shows a watch displayed for sale.

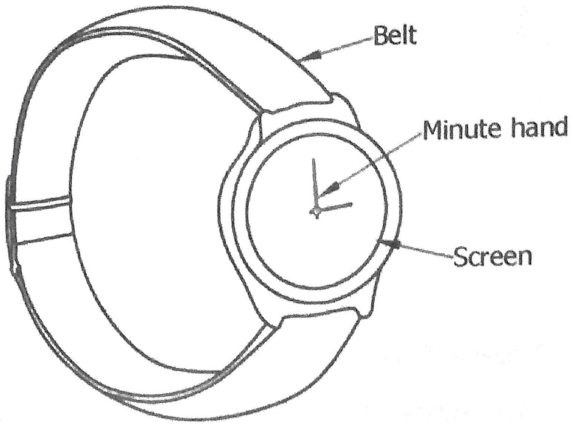



**Figure 9**

**(a)** **(i)** Suggest **two** suitable materials from which the belt could be made. [2]

**(ii)** Give **one** reason for the choice of each material suggested in **(a)(i)**. [2]

**(b)** **Figure 10** shows a circuit diagram of an automatic gate showing component **A**, the relay coil and the L.D. R.

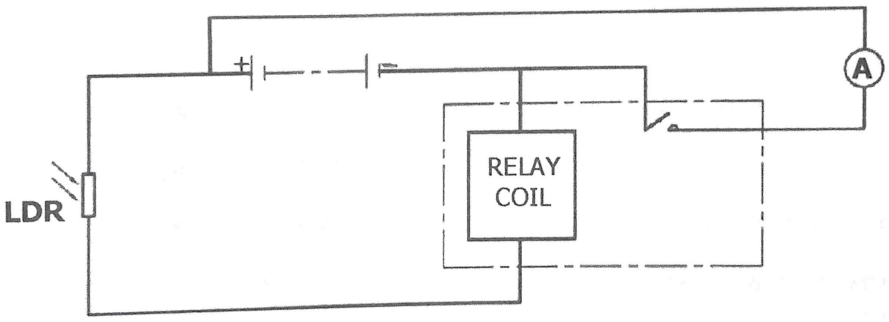



**Figure 10**

**(i)** Identify component **A** and state its function. [2]

**(ii)** What do the letters **L.D. R** stand for? [1]

**(iii)** Describe how the **L. D. R** helps component **A** to operate. [2]

**(c)** Chisels are useful when working on metals and wood.

**(i)** Name **one** type of chisel used on:

- Metal [1]
- Wood [1]

**(ii)** Explain the specific function for each chisel named in **(c)(i)** [2]

**(iii)** Sketch **one** chisel for cutting wood and another **one** for cutting metals. [6]

| Centre Number | Examination Number |
|---|---|
| | |

**(d)** **Figure 11** shows gear used to transmit motion. Gear **A** is connected to the motor.

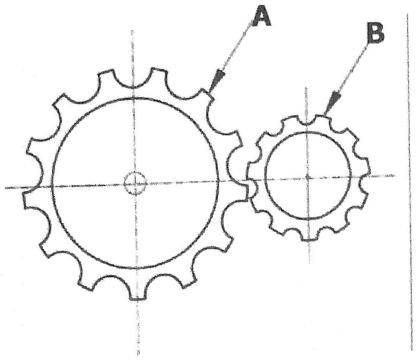



**Figure 11**

**(i)** Suggest **two** specific materials from which gears are made. [2]

**(ii)** Give reasons for using the materials you have stated in **(d)(i)**. [2]

**(iii)** Write **two** functions of linkages in mechanical systems. [2]

**[Total: 25 marks]**

**8** **(a)** **Figure 12** shows a saw used to cut along the grain of wood.



**Figure 12**

**(i)** State the reason why the teeth of the saw in **Figure 12** are 'set'. [1]

**(ii)** Describe how the saw is designed to cut along the grain. [3]

**(iii)** Explain how you would take care of the saw in a workshop. [2]

**(b)** **Figure 13** shows different forces being exerted on the materials.

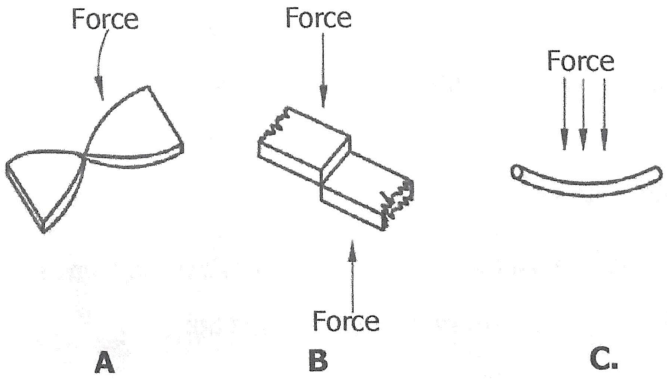



**Figure 13**

**(i)** Name the forces exerted on materials **A**, **B**, and **C**. [3]

**(ii)** With the help of sketches, explain how a metallic tank stand can be strengthened. [5]

**(c)** **Figure 14** shows a process used to produce various products made from plastics.

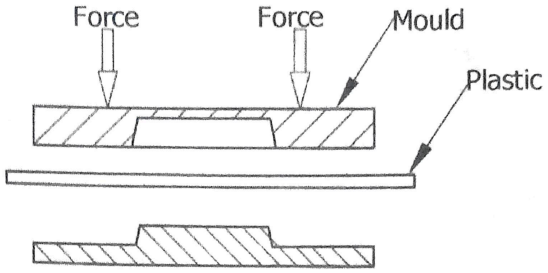


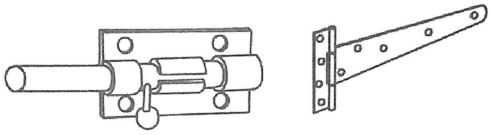

**Figure 14**

**(i)** Identify the method shown in **Figure 14**. [1]

**(ii)** Mention **two** products that are manufactured using the method in **Figure 14**. [2]

**(iii)** State **two** other plastic forming processes. [2]

**(d)** **Figure 15** shows **two** different types of fittings.



**A** **B**

**Figure 15**

**(i)** Identify fittings **A** and **B** [2]

**(ii)** Mention the specific situations where each of the fittings can be used:

fitting **A** [1]

fitting **B** [1]

**(ii)** Differentiate a door lever handle from a cupboard knob. [2]

**[Total: 25 marks]**

**9** **(a)** Sketch the following tools in pictorial view:

**(i)** Tin Snip [3]

**(ii)** Pincers [3]

**(b)** **Table 2** shows items made from different materials.

**(i)** Write a property of each material that makes it suitable for the item mentioned.

| Item | Material | Property |
|---|---|---|
| External Doors | Mukwa | (i) |
| Window Frames | Mild steel | (ii) |
| Packaging item (Television, Radio) | Expanded Polystyrene | (iii) |

[3]

**Table 2**

**(ii)** Sketch a suitable joint you would use on the external door. [3]

| Centre Number | Examination Number |
|---|---|
| | |

**(c)** Outline **three** sources of plastics. [3]

**(d)** **Figure 16** shows one of equipment used as a method of generating power.

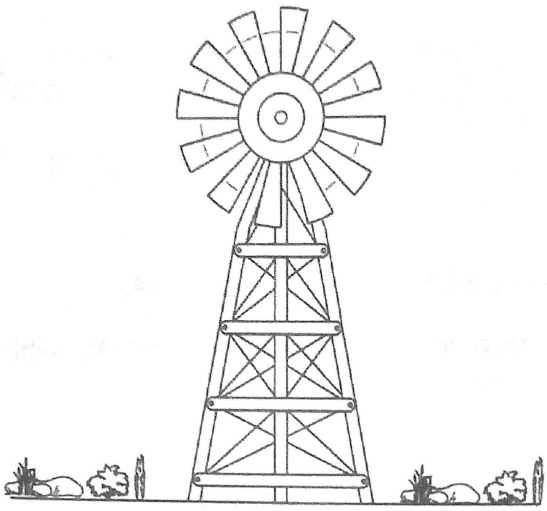


**Figure 16**

**(i)** Why should the frame of equipment in **Figure 16** be made wider at the base? [2]

**(ii)** Give **one** advantage and **one** disadvantage of generating power using the method in **Figure 16**. [2]

**(e)** **Figure 17** shows the pulley system of a bench drilling machine.

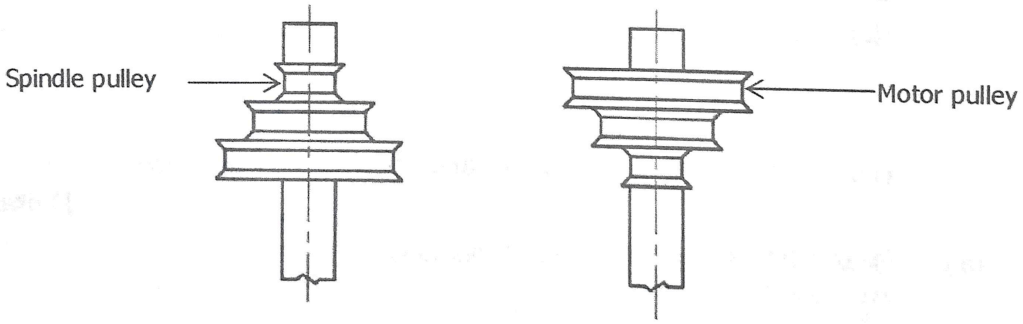



**Figure 17**

**(i)** State **two** uses of the pulley system. [2]

**(ii)** With the help of sketches, show how one can obtain the highest speed of the drilling machine. [3]

**(ii)** Why is it important to change the speed of a drilling machine? [1]

**[Total: 25 marks]**

**Part III: Answer one question on the A3 paper provided.**

**10** **Figure 18** shows a shield without a base.

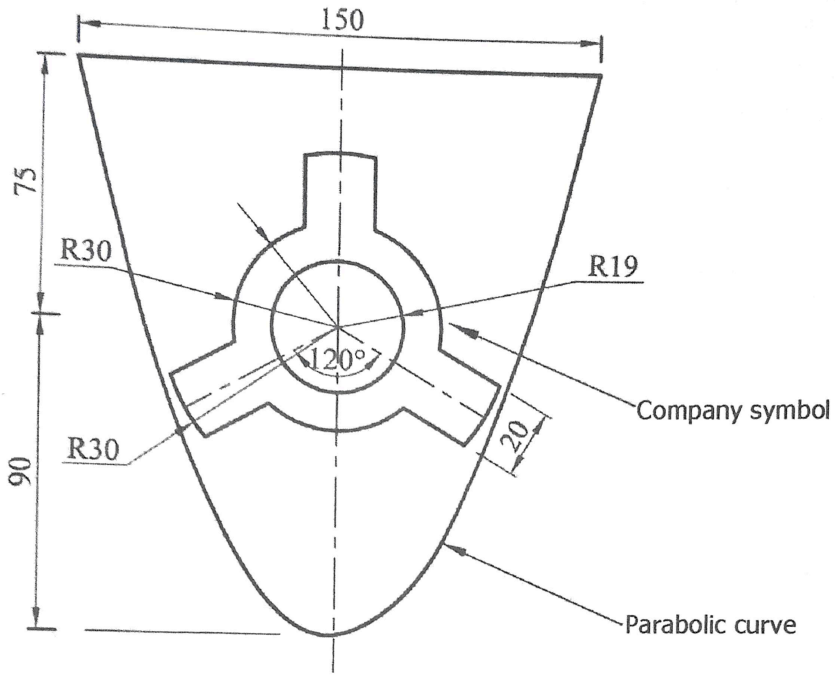



**Figure 18**

**(a)** **(i)** Construct the parabolic profile shown in **Figure 18**. [12]

**(ii)** Construct the symbol engraved on the shield. [11]

**(b)** **(i)** Design a suitable base to be attached to the shield. [4]

**(ii)** Render the shield together with the base. [3]

**[Total: 30 marks]**

| Centre Number | Examination Number |
|---|---|
| | |

**11** **Figure 19** shows an elevation of a truncated cone made from thin hollow Aluminium sheet.

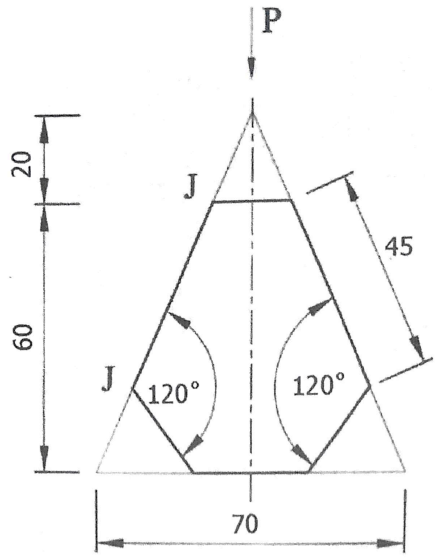



**Figure 19**

**(a)** Construct the given elevation. [14]

**(b)** Construct the surface development of the truncated cone opened along **J – J**. [14]

**(c)** Render the surface development drawn in **(b)**. [2]

**[Total: 30 marks]**